

MOLECULAR DOCKING AND ADME ANALYSIS

Molecular Docking Using SwissDock

Molecular docking studies were performed to evaluate the interaction between selected phytocompounds and target proteins associated with cervical cancer. The three-dimensional structures of Bcl-2 and HPV E6 proteins were obtained from the Protein Data Bank. Protein structures were prepared by removing heteroatoms and selecting the appropriate chain. Ligand structures (2,4-di-tert-butylphenol, tetradecanal, and pyridine-3-carboxamide) were retrieved from the PubChem database. Docking was carried out using the SwissDock online server under default parameters. Binding affinities and interaction patterns were analyzed based on docking scores

ADME Analysis

ADME properties of selected compounds were evaluated using the SwissADME online tool. Ligand structures obtained from PubChem were uploaded to the server. Parameters such as absorption, distribution, metabolism, excretion, and drug-likeness were analyzed to assess their suitability as potential drug candidates

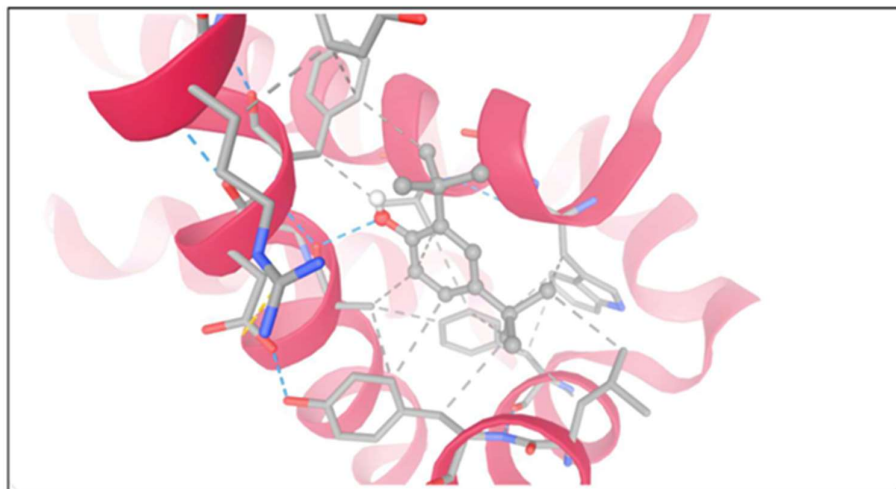
RESULTS

MOLECULAR DOCKING

Molecular docking results

S. No	Ligand	Target Protein	Binding Affinity (kcal/mol)
1	2,4-di-tert-butylphenol	HPV E6	-5.423
2	2,4-di-tert-butylphenol	Bcl-2	-6.010
3	Tetradecanal	HPV E6	-4.187
4	Tetradecanal	Bcl-2	-4.772
5	Pyridine-3-carboxamide	HPV E6	-4.236
6	Pyridine-3-carboxamide	Bcl-2	-4.711

The highest binding affinity was observed for 2,4-di-tert-butylphenol with Bcl-2 (−6.01 kcal/mol), suggesting a relatively stable interaction



Docking position interaction between 2,4-di-tert-butylphenol and Bcl-2 protein

ADME Analysis

Property	2,4-di-tertbutylphenol	Tetradecanal	Pyrimidine-3-carboxamide
Lipinski Rule	Yes; 0 violations	Yes; 0 violations	Yes; 0 violations
GI Absorption	High	High	High
Bioavailability Score	0.55	0.55	0.55
Drug-likeness	High	Moderate	Moderate

Among the compounds, 2,4-di-tert-butylphenol exhibited the most favourable drug-likeness, indicating its suitability as a potential drug candidate